SMARTBRIDGE
Let's Bridge the Gap

SKILLWALLET
A SmartBridge Product

Technology Stack

Date	15 March 2026
Team ID	SWTID-2026-6775
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	React / Next.js, Tailwind CSS	Provides a highly responsive, component-based user interface for dynamic form submission, visual workout preview, and real-time user interactive dashboards.
2	**Backend / Server-Side** *(e.g., Node.js, Python/FastAPI)*	Python (FastAPI / Flask)	Lightweight, efficient, and seamlessly integrates with Google Gemini SDKs to process user metrics and orchestrate prompt workflows.
3	**AI / LLM Integration** *(e.g., Gemini API, LangChain)*	Google Gemini Models (Gemini 1.5 Pro / Flash)	Generates customized, context-aware workout and nutrition recommendations based on user targets, dietary preferences, and fitness levels.
4	**Database Storage** *(e.g., PostgreSQL, MongoDB)*	MongoDB / Firebase Firestore	Provides flexible, document-oriented storage to manage user profiles, generated fitness plans, and activity history efficiently.

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2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	*[Specify technology]*	*[Explain why this was chosen (e.g., fast processing, existing team expertise)]*
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	*[Specify technology]*	*[Explain why this was chosen (e.g., handles unstructured data, strong relational mapping)]*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*[Specify technology]*	*[Explain why this was chosen (e.g., easy scalability, cost-effective hosting)]*
5	**Version Control & CI/CD**	*[Specify technology]*	*[Explain why this was chosen (e.g., seamless collaboration, automated testing)]*

	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*[Specify technology]*	*[Explain the specific feature this enables in your project]*